

Hawkeye EQMS · CAPA v2 — Corrective & Preventive Action · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

Document version	v1.1
Generated on	2026-04-26T00:05:47.121Z
Tenant under test	Novex Pharma Inc. · 69e64e7869b2ba745d40bb89
Live frontend	https://hawkeye-frontend-dev-chi.vercel.app
Live backend	https://hawkeye-backend-dev.vercel.app

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1. Module overview

Module	CAPA v2 — Corrective & Preventive Action
Pharma purpose	Formalised corrective + preventive action lifecycle. Intake → triage → investigation → RCA → action plan → implementation → effectiveness review → closure. 5 mandatory approval gates.
Compliance basis	ISO 9001:2015 §10.2 · 21 CFR 211.192 · 21 CFR 820.100 (CAPA · medical device) · ICH Q10 §3.2.2 · EU GMP Chapter 1.4
Hawkeye routes	/buyer/capas, /auditor/capas (frontend) · /api/capa-v2/* (backend)
Hawkeye model file	backend/src/models/capaV2Models.js (17 collections)
Tenant module flag	modules.CAPA

CAPA v2 supersedes the legacy single-collection capas with a 17-collection event-sourced model: every state transition writes a capa-v2-status-history row, every approval writes a capa-v2-approvals row, every action item is its own document, and every implementation artefact has its own capa-v2-implementation-evidence row. This makes it possible to satisfy 21 CFR Part 11 audit-trail requirements without separate logging — the data IS the audit trail. Every state transition is gated by the statusMachine.js state graph (no skipping states without admin override) and by role-based permits.

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. **MET** = parity, **PARTIAL** = ships but with caveats, **GAP** = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
5-stage approval workflow (Triage / RCA / Action Plan / Effectiveness / Closure) 21 CFR 820.100	POST /api/capa-v2/capas/id/approvals with stage enum + decision (APPROVED/REJECTED/NEEDS_REWORK). Per-stage gating in statusMachine.js.	MET	

Pharma expectation	Hawkeye implementation	Status	Note
Auto-generated CAPA number (CAPA-YYYY-NNNN) GMP traceability	nextCapaNumber() in modules/capaV2/prefillService.js. Per-tenant sequence with year prefix.	MET	
Multiple sources: audit finding, deviation, complaint, regulator inspection ICH Q10 §3.2.5	sourceType enum on intake + capa records: AUDIT_FINDING / DEVIATION / COMPLAINT / INSPECTION / INTERNAL / CUSTOMER_FEEDBACK. capa-v2-source-links table for traceability.	MET	
Triage decision: Formal CAPA / Correction Only / No CAPA Needed ICH Q10 §3.2.2	POST /triage/:id/decision with decision enum + rationale. Drives next state.	MET	
Structured RCA capture (5-Why + Fishbone + FMEA + verification) ICH Q9	PUT /capas/:id/root-cause with rcaMethod enum + rootCauseStatement + verification. Stored in capa-v2-root-causes.	MET	
Action plan with corrective + preventive + interim actions, each with owner + due date + status 21 CFR 820.100(a)	PUT /capas/:id/action-plan + POST /capas/:id/action-items. Each action-item is its own document with status NOT_STARTED → IN_PROGRESS → COMPLETED.	MET	
Implementation evidence (uploaded documents, test results) 21 CFR 211.192	POST /capas/:id/implementation-evidence — accepts S3 refs + metadata. Stored in capa-v2-implementation-evidence.	MET	
Effectiveness check after implementation (PASS / FAIL) 21 CFR 820.100(a)(4)	PUT /capas/:id/effectiveness with result enum + reviewWindow. Stored in capa-v2-effectiveness-checks.	MET	
Closure: CLOSED_EFFECTIVE or CLOSED_INEFFECTIVE 21 CFR 820.100(a)(5)	POST /capas/:id/close with outcome (EFFECTIVE / INEFFECTIVE) + closureReason. Final approval gate.	MET	
Re-open if recurrence detected ICH Q10 §3.2.4	POST /capas/:id/reopen with reason. Status flips to REOPENED.	MET	
Risk assessment captured (patient / product / compliance / recurrence) ICH Q9	POST /capas/:id/risk-assessment. Stored in capa-v2-risk-assessments.	MET	
Audit trail of every state transition (ALCOA+) 21 CFR Part 11 §11.10(e)	Every transitionCapaStatus() call writes a capa-v2-status-history row with from/to/userId/reason/timestamp. Mongoose pre-save hook.	MET	
Daily metric snapshot (age, overdue, recurrence) QMS KPI	createMetricSnapshot() called on close + reopen. capa-v2-metric-snapshots collection.	MET	
Similar / recurrence / duplicate detection ICH Q10 §3.2.4	capa-v2-similarity-links table with linkType (SIMILAR / RECURRENCE / POTENTIAL_DUPLICATE). Population currently manual.	PARTIAL	AI-driven similarity scoring on roadmap
Cross-module CAPA creation from deviation/complaint/finding via single button —	POST /api/quality/capa-from-deviation creates a real v2 record + action plan + links. Wired to AI Draft RCA drawer.	MET	
AI: predictive on-time / effectiveness scoring FDA AI guidance Jan 2025	POST /api/ai/predict/capa-outcome (Wave 3 heuristic). Returns P(on-time) + P(effective) + top factors.	MET	

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026` . Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()` .

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Audit Lead Maria auditor	audit.lead@novex-pharma.demo	Reviews intake, makes triage decision, approves RCA + action plan + closure. Owns the approval gates.	UNDER_TRIAGE RCA_PENDING_APPROVAL ACTION_PLAN_PENDING_APPROVAL EFFECTIVENESS_REVIEW CLOSED
QA Specialist Kenji admin	qa.specialist@novex-pharma.demo	Submits intakes from deviations / complaints, executes investigation + RCA + action items, uploads evidence.	INTAKE_DRAFT INVESTIGATION_IN_PROGRESS IN_IMPLEMENTATION AWAITING_EFFECTIVENESS_CHECK
Head of QA James admin	qa.head@novex-pharma.demo	Owns CAPAs end-to-end across the tenant; assigns ownership; signs off effectiveness.	CAPA_OPEN onwards
VP Quality Elena tenant_admin	vp.quality@novex-pharma.demo	Final closure approval on CRITICAL CAPAs; reviews CAPA backlog at MRM.	CLOSED_EFFECTIVE CLOSED_INEFFECTIVE

4. Feature catalogue (click-by-click)

4.1 CAPA workspace dashboard

What it does	Counts of open / closed-effective / closed-ineffective / overdue CAPAs + bySeverity breakdown + overdue action-items count.
Menu / route	/buyer/capas (admin/buyer view) or /auditor/capas (auditor view)
Available to roles	admin buyer auditor tenant_admin
Backend API	GET /api/capa-v2/dashboard

CLICK-BY-CLICK

- NAVIGATE Click 'CAPAs' in the top nav
→ Dashboard renders with counter cards
- WAIT Spinner clears (~700 ms)
→ counts.open · counts.closedEffective · counts.overdue · bySeverity render

state-screens/capa-buyer.png

What it does	Promote an audit finding to a CAPA candidate so it's tracked in the workspace before formal intake.
Menu / route	Audit detail · finding row · 'Promote to CAPA candidate' button (or POST API)
Available to roles	auditor admin tenant_admin
Backend API	POST /api/capa-v2/candidates/from-finding

1. **CLICK** On an audit finding, click 'Promote to CAPA candidate'
→ capa-v2-candidates row created with status NEW; sourceType=AUDIT_FINDING

What it does	One-click — every Major+ finding from a closed audit becomes a CAPA candidate.
Menu / route	Audit closure step · 'Generate CAPAs from findings' button
Available to roles	auditor tenant_admin

Backend API

POST /api/capa-v2/candidates/bulk-from-audit/:auditId

CLICK-BY-CLICK

1. **CLICK** On the audit closure page, click 'Generate CAPAs'
→ One candidate per finding above the configured severity threshold

4.4 Create + submit intake

What it does

Author the intake form (issueStatement, scope, regulatoryExposure). Submit moves the candidate into UNDER_TRIAGE.

Menu / route

/buyer/capas → 'New CAPA intake' drawer

Available to roles

admin

buyer

auditor

Backend API

POST /api/capa-v2/intakes · POST /api/capa-v2/intakes/:id/submit

CLICK-BY-CLICK

1. **CLICK** Click 'New CAPA intake'
→ Drawer opens
2. **TYPE** Fill issueStatement (1 sentence)
→ required
3. **TYPE** Fill issueDescription (full narrative)
→ required
4. **CLICK** Choose severity (CRITICAL / MAJOR / MINOR)
→ default MINOR
5. **CLICK** Click 'Save' (status=DRAFT)
→ Intake row appears with state INTAKE_DRAFT
6. **CLICK** Click 'Submit for triage'
→ Status flips to UNDER_TRIAGE; triage row created

FIELDS CAPTURED

Field	Required	Validation / values	Notes
issueStatement	Yes	string	
issueDescription	Yes	multi-line text	
severity	No	CRITICAL MAJOR MINOR	default MINOR
sourceType	No	AUDIT_FINDING DEVIATION COMPLAINT INTERNAL INSPECTION	auto-set when promoted from a source record

4.5 Triage decision

What it does

Triage reviewer (auditor) decides FORMAL_CAPA / CORRECTION_ONLY / NO_CAPA_NEEDED.

Menu / route

Triage row · 'Triage decision' button

Available to roles

auditor

tenant_admin

admin

superadmin

Backend API

POST /api/capa-v2/triage/:id/decision

CLICK-BY-CLICK

1. **CLICK** Click 'Triage decision'
→ Drawer opens
2. **CLICK** Choose decision: FORMAL_CAPA_REQUIRED / CORRECTION_ONLY / NO_CAPA_NEEDED
→ Drives the next state
3. **TYPE** Fill rationale
→ required
4. **CLICK** Click 'Submit decision'

→ If FORMAL_CAPA → CAPA_OPEN; if CORRECTION_ONLY → IN_IMPLEMENTATION; if NO_CAPA → TRIAGE_NO_CAPA terminal

4.6 Open CAPA + assign owner

What it does	Mint the CAPA-v2 record once triage decides FORMAL_CAPA. Owner + due date required.
Menu / route	Triage decision flow · 'Create CAPA' button
Available to roles	triageRoles
Backend API	POST /api/capa-v2/capas

CLICK-BY-CLICK

1. [CLICK](#) Click 'Create CAPA' on the triage drawer
→ Drawer expands
2. [CLICK](#) Pick owner from user dropdown
→ ownerId required
3. [CLICK](#) Set dueDate (default +30 days)
→ required
4. [CLICK](#) Click 'Open CAPA'
→ Status=CAPA_OPEN; capaNumber generated; auto-redirect to detail

4.7 Save investigation

What it does	Investigator records investigation scope + summary + findings.
Menu / route	CAPA detail · Investigation tab
Available to roles	manageRoles
Backend API	PUT /api/capa-v2/capas/:id/investigation

CLICK-BY-CLICK

1. [CLICK](#) Open Investigation tab
→ Form renders
2. [TYPE](#) Fill investigationScope, summary, findings, evidenceLinks[]
→ All optional but recommended
3. [CLICK](#) Click 'Save investigation'
→ Status moves to INVESTIGATION_IN_PROGRESS

4.8 Save root-cause analysis

What it does	Capture the structured RCA — method enum + statement + verification.
Menu / route	CAPA detail · RCA tab
Available to roles	manageRoles
Backend API	PUT /api/capa-v2/capas/:id/root-cause
AI assistance	AI AI 5-Why scaffolder (Wave 1 · POST /api/ai/deviation/scaffold-five-why) can pre-populate the 5 whys

CLICK-BY-CLICK

1. [CLICK](#) Open RCA tab
→ Form renders
2. [CLICK](#) Choose rcaMethod (FIVE_WHY / FISHBONE / FMEA / FAULT_TREE / PARETO)
→ required
3. [TYPE](#) Fill rootCauseStatement
→ required
4. [CLICK](#) Click 'Submit for approval'

→ Status moves to RCA_PENDING_APPROVAL

4.9 Approve stage (RCA / Action Plan / Effectiveness / Closure)

What it does	Approver reviews the stage submission and APPROVES / REJECTS / NEEDS_REWORK.
Menu / route	CAPA detail · 'Approve' button when status is *_PENDING_APPROVAL
Available to roles	auditor tenant_admin admin superadmin
Backend API	POST /api/capa-v2/capas/:id/approvals

CLICK-BY-CLICK

1. [CLICK](#) Click 'Approve' button
→ Drawer with stage label
2. [CLICK](#) Choose decision (APPROVED / REJECTED / NEEDS_REWORK)
→ required
3. [TYPE](#) Fill comments + (optional) e-signature meaning
→ comments required
4. [CLICK](#) Click 'Submit decision'
→ Status advances to next state per state machine

4.10 Save action plan + add action items

What it does	Capture corrective + preventive + interim actions. Then add granular action-items each with owner + dueDate.
Menu / route	CAPA detail · Action Plan tab
Available to roles	manageRoles
Backend API	PUT /api/capa-v2/capas/:id/action-plan · POST /api/capa-v2/capas/:id/action-items

CLICK-BY-CLICK

1. [CLICK](#) Open Action Plan tab
→ Plan summary form
2. [TYPE](#) Fill correctiveActionsSummary + preventiveActionsSummary
→ AI Draft RCA pre-populates these if invoked from a deviation
3. [CLICK](#) Click '+ Add action item'
→ Item form
4. [TYPE](#) Fill description + assignedToUserId + dueDate
→ All required
5. [CLICK](#) Click 'Submit for approval'
→ Status moves to ACTION_PLAN_PENDING_APPROVAL

4.11 Update action-item status (start / complete / block)

What it does	Action-item owner moves their item through NOT_STARTED → IN_PROGRESS → COMPLETED (or BLOCKED).
Menu / route	Action item card · status dropdown
Available to roles	manageRoles
Backend API	PATCH /api/capa-v2/action-items/:id/status

CLICK-BY-CLICK

1. [CLICK](#) Click status dropdown on an action-item card
→ Status enum shown
2. [CLICK](#) Pick IN_PROGRESS or COMPLETED
→ Item updates; CAPA status moves to IN_IMPLEMENTATION when first item starts

4.12 Upload implementation evidence

What it does	Attach documents proving the action item was completed (test results, updated SOP, training records).
Menu / route	CAPA detail · Implementation tab · '+ Upload evidence' button
Available to roles	manageRoles
Backend API	POST /api/capa-v2/capas/:id/implementation-evidence

CLICK-BY-CLICK

1. [CLICK](#) Click '+ Upload evidence'
→ Drawer with file picker + metadata fields
2. [CLICK](#) Pick file (uploads to S3) or paste link
→ documentRef captured
3. [TYPE](#) Fill description + actionItemId link
→ required
4. [CLICK](#) Click 'Save evidence'
→ evidence row appears under the action item

4.13 Begin + record effectiveness check

What it does	After all actions COMPLETED + grace period passed, run the effectiveness review.
Menu / route	CAPA detail · Effectiveness tab
Available to roles	manageRoles
Backend API	PUT /api/capa-v2/capas/:id/effectiveness

CLICK-BY-CLICK

1. [CLICK](#) Click 'Begin effectiveness review'
→ Status moves to EFFECTIVENESS_REVIEW_IN_PROGRESS
2. [TYPE](#) Fill review window dates + observations
→ Required
3. [CLICK](#) Choose result (PASS / FAIL)
→ Drives close outcome

4.14 Close CAPA

What it does	Final closure. Captures closureOutcome (EFFECTIVE / INEFFECTIVE) + reason. Approval-gated.
Menu / route	CAPA detail · 'Close CAPA' button
Available to roles	approvalRoles
Backend API	POST /api/capa-v2/capas/:id/close

CLICK-BY-CLICK

1. [CLICK](#) Click 'Close CAPA'
→ Confirmation dialog
2. [CLICK](#) Pick outcome (EFFECTIVE / INEFFECTIVE)
→ required
3. [TYPE](#) Fill closureReason
→ required
4. [CLICK](#) Click 'Confirm + sign'
→ Status flips to CLOSED_EFFECTIVE / CLOSED_INEFFECTIVE; metric snapshot written

4.15 Predictive CAPA badge (Wave 3)

What it does	Inline P(on-time) + P(effective) + top factors.
Menu / route	CAPA row badge + drawer card
Available to roles	any viewer
Backend API	POST /api/ai/predict/capa-outcome
AI assistance	AI Heuristic model capa.heuristic@1.0.0 — features: severity, rootCauseDepth, actionCount, daysToDue

CLICK-BY-CLICK

1. WAIT Auto-renders on row open
- Two chips: P(on-time) and P(effective)

5. Full lifecycle walkthrough — one transaction, end-to-end

One CAPA walked from candidate intake through every approval gate to CLOSED_EFFECTIVE. Source = the deviation lifecycle from the Deviation guide.

#	Persona	State transition	Action	API	Observed	Result
1	Kenji QA Specialist	– → INTAKE_DRAFT	From the deviation View+AI drawer, click 'Create CAPA from this RCA' (cross-module). Cross-module endpoint mints a real CAPA + intake.	POST /api/quality/capa-from-deviation	capaNumber=CAPA-2026-NNNN · status=CAPA_OPEN · sourceType=DEVIATION · linked back to deviation.linkedCAPAIds	PASS

Expected DB state after this step:

```
capa-v2 { _id, capaNumber: 'CAPA-2026-NNNN', status: 'CAPA_OPEN', sourceType: 'DEVIATION', sourceRef: { id, collection: 'deviations', label: 'DEV-NNNN' }, ownerUserId, dueDate: +30d, createdBy }
capa-v2-status-history { from: null, to: 'CAPA_OPEN', userId, ts }
capa-v2-action-plans { capaId, correctiveActionsSummary, preventiveActionsSummary }
```

#PersonaState transitionActionAPIObservedResult

Audits

New Audit

CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

Buyer > CAPAs

CAPA Management Workspace

OPEN5

OVERDUE0

RECURRING0

NEW CANDIDATES0

Audit ID

Generate

Manual CAPA title

Create

Search candidates

Candidate StatusAll

You don't have any capa candidate queue

Search CAPAs

CAPA StatusAll

Refresh

Q. Fuzzy search

DensityStandard

View

Save As

Save View

Reset

Manage

Export to CSV

Manage Columns

CAPA ID	Title	Status	Severity	Due Date	Owner
CAPA-2026-00005	CAPA from DEV-2026-0064: F1 X-mod - OOS dissolu...	CAPA_OPEN	HIGH	5/25/2026, 11:05:21 ...	supplier_quality_lead
CAPA-2026-00004	CAPA from DEV-2026-0063: F1 X-mod - OOS dissolu...	CAPA_OPEN	HIGH	5/25/2026, 11:02:31 ...	supplier_quality_lead
CAPA-2026-00003	CAPA from DEV-2026-0062: F1 X-mod - OOS dissolu...	CAPA_OPEN	HIGH	5/25/2026, 10:59:15 ...	supplier_quality_lead
CAPA-2026-00002	CAPA from DEV-2026-0061: F1 X-mod - OOS dissolu...	CAPA_OPEN	HIGH	5/25/2026, 10:54:59...	supplier_quality_lead
CAPA-2026-00001	CAPA from DEV-2026-0060: P	CAPA_OPEN	HIGH	5/25/2026, 10:54:13 ...	supplier_quality_lead

CAPA Detail

OverviewInvestigationAction PlanEffectivenessHistory

CAPA-2026-00005 | CAPA_OPEN

CAPA from DEV-2026-0064: F1 X-mod - OOS dissolution on demo batch

Owner User ID69e64e7a69b2ba745d40bb6

Owner Rolesupplier_quality_lead

Assign

StageRCA

DecisionAPPROVED

Approval Note

Approve

Comment

Add Comment

Close Effective

Reopen

state-screens/capa-buyer.png

2	Maria Lead Auditor	CAPA_OPEN → INVESTIGATION_IN_PROGRESS	Open CAPA detail → Investigation tab → fill scope + findings → Save investigation	PUT /api/capa-v2/capas/:id/investigation	Status moves; capa-v2-investigations row written	PASS
Expected DB state after this step: <pre>capa-v2-investigations { capaId, scope, summary, findings, evidenceLinks: [] } capa-v2-status-history { from: 'CAPA_OPEN', to: 'INVESTIGATION_IN_PROGRESS' }</pre>						
3	Maria Lead Auditor	INVESTIGATION_IN_PROGRESS → RCA_PENDING_APPROVAL	RCA tab → method=FIVE_WHY → fill rootCauseStatement (paste from AI 5-Why output) → Submit	PUT /api/capa-v2/capas/:id/root-cause	Status flips; capa-v2-root-causes row written	PASS

#	Persona	State transition	Action	API	Observed	Result
Expected DB state after this step: <pre> capa-v2-root-causes { capaId, rcaMethod: 'FIVE_WHY', rootCauseStatement, verification: {} } capa-v2-status-history { from: 'INVESTIGATION_IN_PROGRESS', to: 'RCA_PENDING_APPROVAL' } </pre>						
4	James Approver (admin)	RCA_PENDING_APPROVAL → ACTION_PLAN_PENDING_APPROVAL	Click 'Approve RCA' → decision=APPROVED → comments → Submit	POST /api/capa-v2/capas/:id/approvals (stage=RCA, decision=APPROVED)	Status advances; capa-v2-approvals row written	PASS
Expected DB state after this step: <pre> capa-v2-approvals { capaId, stage: 'RCA', decision: 'APPROVED', userId, comments, decidedAt } </pre>						
5	Kenji QA Specialist	ACTION_PLAN_PENDING_APPROVAL → ACTION_PLAN_APPROVED	(Already submitted via Create-CAPA; James approves it). Approve action plan.	POST /api/capa-v2/capas/:id/approvals (stage=ACTION_PLAN, decision=APPROVED)	Status moves; one capa-v2-approvals row added	PASS
6	Action-item owners varied	ACTION_PLAN_APPROVED → IN_IMPLEMENTATION	Each owner moves their action-item NOT_STARTED → IN_PROGRESS	PATCH /api/capa-v2/action-items/:id/status	First IN_PROGRESS auto-promotes parent CAPA to IN_IMPLEMENTATION	PASS
Expected DB state after this step: <pre> capa-v2-action-items[*].status='IN_PROGRESS' </pre>						
7	Action-item owners varied	IN_IMPLEMENTATION → AWAITING_EFFECTIVENESS_CHECK	All action-items → COMPLETED + at least one capa-v2-implementation-evidence row uploaded	PATCH /api/capa-v2/action-items/:id/status · POST /api/capa-v2/capas/:id/implementation-evidence	Status moves; evidence rows visible	PASS
Expected DB state after this step: <pre> All capa-v2-action-items.status='COMPLETED' capa-v2-implementation-evidence[*] </pre>						
8	Maria Effectiveness reviewer	AWAITING_EFFECTIVENESS_CHECK → EFFECTIVENESS_REVIEW_IN_PROGRESS	Effectiveness tab → Begin review → reviewWindow + observations → Save	PUT /api/capa-v2/capas/:id/effectiveness	capa-v2-effectiveness-checks row created with result=PENDING	PASS
Expected DB state after this step: <pre> capa-v2-effectiveness-checks { capaId, result: 'PENDING', reviewWindowStart, reviewWindowEnd } </pre>						
9	Elena VP Quality	EFFECTIVENESS_REVIEW_IN_PROGRESS → CLOSED_EFFECTIVE	Click 'Close CAPA' → outcome=EFFECTIVE → closureReason → Confirm	POST /api/capa-v2/capas/:id/close	Status=CLOSED_EFFECTIVE · closedAt set · capa-v2-metric-snapshots row written	PASS
Expected DB state after this step: <pre> capa-v2 { status: 'CLOSED_EFFECTIVE', closedAt, closureOutcome: 'EFFECTIVE' } capa-v2-metric-snapshots { capaId, ageDays, actionItemsClosed, result: 'EFFECTIVE' } </pre>						

6. AI assistance map

Each AI assist is a separate endpoint. Outputs flow through the grounded-generation runtime (citation gate, confidence floor, schema validation, accept/reject audit trail).

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
AI 5-Why Scaffolder (Wave 1)	INVESTIGATION_IN_PROGRESS	POST /api/ai/deviation/scaffold-five-why	RCA tab · 'Scaffold 5-Why with AI' button	5-level Why chain with citations + 6 follow-up questions + 6M categorisation	Free Gemini 2.5 Flash-Lite
AI CAPA RCA Drafter (Wave 1)	INVESTIGATION_IN_PROGRESS → ACTION_PLAN_PENDING_APPROVAL	POST /api/ai/capa/draft-rca	Pre-populates RCA + Action Plan tabs when invoked from a deviation	Full RCA + corrective + preventive + effectiveness draft	Free Gemini 2.5 Flash-Lite
Predictive CAPA Outcome (Wave 3)	any post-CAPA_OPEN	POST /api/ai/predict/capa-outcome	CAPA row badge + detail card	P(on-time) + P(effective) + top contributing factors	Heuristic capa.heuristic@1.0.0
Active Learning Loop	CLOSED_EFFECTIVE CLOSED_INEFFECTIVE	POST /api/ai/active-learning/adjustments	Drift dashboard	Refines AI retrieval-set weights based on accept/reject outcomes	Wave 3 retrieval re-weighting
(roadmap) AI similarity scoring	CAPA_OPEN, CLOSED_*	(not wired)	(future)	Auto-populate capa-v2-similarity-links with SIMILAR/RECURRENCE links from embedding similarity	Vector DB (pgvector planned)

7. Regulator-trace matrix

For each lifecycle state, the specific clauses + evidence Hawkeye captures so an inspector can audit the record. Maps to FDA / ISO / ICH / EU GMP source clauses.

State	Cited clauses	Evidence captured	Backing records
INTAKE_DRAFT	21 CFR 820.100(a)(1) ISO 9001 §10.2.1	issueStatement + issueDescription + sourceType captured	capa-v2-intakes
UNDER_TRIAGE	ICH Q10 §3.2.2	Triage decision rationale + decider + timestamp	capa-v2-triage
INVESTIGATION_IN_PROGRESS	21 CFR 211.192 21 CFR 820.100(a)(2)	Investigation scope + findings + evidenceLinks	capa-v2-investigations
RCA_PENDING_APPROVAL	ICH Q9 §IV 21 CFR 820.100(a)(3)	rcaMethod + rootCauseStatement + verification	capa-v2-root-causes + capa-v2-approvals (stage=RCA)
ACTION_PLAN_APPROVED	21 CFR 820.100(a)(4) ISO 9001 §10.2.1(d)	correctiveActions + preventiveActions + per-item owner + dueDate	capa-v2-action-plans + capa-v2-action-items + capa-v2-approvals (stage=ACTION_PLAN)
IN_IMPLEMENTATION	21 CFR 820.100(a)(4)	Each completed action-item + uploaded evidence (S3 ref + metadata)	capa-v2-action-items + capa-v2-implementation-evidence
EFFECTIVENESS_REVIEW	21 CFR 820.100(a)(4) ICH Q10 §3.2.4	Review window dates + observations + result (PASS / FAIL)	capa-v2-effectiveness-checks
CLOSED_EFFECTIVE	21 CFR 820.100(a)(5) 21 CFR Part 11 §11.50	Closure approval + closedBy + closedAt + closureOutcome + e-sig	capa-v2-approvals (stage=CLOSURE) + capa-v2-status-history (final row)

8. Test results matrix

Suite	Scope	Result	Evidence
eqms-lifecycle.spec.ts · capa-v2	GET workspace · POST intake · POST submit · UI workspace render	PASS	4/4 PASS · eqms-test-results-v2.pdf
eqms-cross-module.spec.ts · F1	Deviation → AI RCA → real CAPA-v2 created in workspace + linked	PASS	F1 steps 3-6 PASS
eqms-cross-module.spec.ts · F3	Complaint → CAPA-v2 intake (sourceType=COMPLAINT) → linked back	PASS	F3 steps 3-4 PASS
Wave 3 smoke	POST /api/ai/predict/capa-outcome	PASS	wave3 smoke

9. Known gaps + roadmap

- **Full UI for the 5 approval stages** — Today the workspace shell is built but the per-tab forms are partially scaffolded. Need full UI for Investigation / RCA / Action Plan / Effectiveness drawers with required-field gating. **HIGH**
- **AI similarity / recurrence detector** — capa-v2-similarity-links table exists but is empty. Plug embedding-based similarity (pgvector) to surface 'this CAPA looks like CAPA-2025-019 — recurrence?' suggestions at intake. **MEDIUM**
- **Action-item dependency graph** — Action-items can have parents but the UI doesn't visualise the DAG. Add a Gantt-style or graph view. **LOW**
- **Effectiveness review reminder scheduler** — When an action plan reviewWindow opens, send a notification to the reviewer (notificationOutbox). **MEDIUM**
- **Bulk-import legacy capas → v2** — Legacy capas collection is read-only. Provide a one-shot migration endpoint that maps old fields to v2 + preserves the audit trail. **LOW**